

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Understand ethical issues in AI-assisted coding
- Identify risks related to privacy, security, and bias
- Analyze transparency in AI-generated code
- Apply responsible coding practices

Task 1: Privacy in API Usage

Problem: Generate Python code to fetch weather data securely.

Insecure Code

```
api_key = "12345ABC"  
print("Fetching weather data using API key:", api_key)
```

Secure Code

```
import os  
  
api_key = os.getenv("API_KEY")  
print("Fetching weather data securely")
```

Analysis: Hardcoding API keys is insecure. Environment variables improve security.

Task 2: Privacy in File Handling

Problem: Store user credentials securely.

Insecure Code

```
file = open("users.txt", "w")
file.write("user,password123")
file.close()
```

Secure Code

```
import hashlib

password = "password123"
hashed = hashlib.sha256(password.encode()).hexdigest()
print("Stored hashed password:", hashed)
```

Analysis: Plain text storage is unsafe. Hashing protects user data.

Task 3: Transparency in Algorithm Design

Python Code

```
num = int(input("Enter number: "))
temp = num
sum = 0
digits = len(str(num))

while temp > 0:
    digit = temp % 10
    sum += digit ** digits
    temp //= 10

if sum == num:
    print("Armstrong")
else:
    print("Not Armstrong")
```

Explanation: Each digit is raised to the power of total digits and summed.

Task 4: Algorithm Comparison

Python Code

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(n - i - 1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
    return arr

def quick_sort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[0]
    left = [x for x in arr[1:] if x <= pivot]
    right = [x for x in arr[1:] if x > pivot]
    return quick_sort(left) + [pivot] + quick_sort(right)
```

Analysis: Bubble Sort is simple but slow. Quick Sort is faster and efficient.

Task 5: Explainable Recommendation System

Python Code

```
interest = input("Enter interest: ")

if "data" in interest:
    print("Recommended: Data Science (based on your interest)")
elif "web" in interest:
    print("Recommended: Web Development (based on your interest)")
else:
    print("Recommended: Programming Basics")
```

Analysis: Providing explanations increases transparency and trust.

Conclusion

Ethical AI coding ensures security, transparency, and fairness.

Developers must carefully review AI-generated code to avoid risks and ensure responsible usage.